Petrographic Observations of the Apollo Lunar Thin Section 60025-r1

Ping Wang

September 2025

1 Introduction

Sample 60025 was collected near the Lunar Module and its orientation was documented by photography [2]. The paired rotational series of thin section 60025-r1, initially downloaded from Virtual Microscope [1] and subsequently aligned by using Python in order to have consistent field of view in PPL and XPL.

2 Petrographic Observations

2.1 Observations in PPL

In PPL, this thin section is characterized by:

- Predominantly colorless and low-relief grains.
- Numerous elongate and tabular grains dispersed throughout the section.
- Absence of strong pleochroism in most minerals.
- A large fractured grain near the center of the field of view exhibiting: irregular polygonal outline, extensive internal cracks, moderate relief relative to surrounding grains, and appearing pale brown.
- Several smaller fractured grains distributed around the central crystal.
- Fine crack networks extending through portions of the matrix.
- Sparse tiny opaque specks occur throughout.

The overall appearance suggests a largely mineral assemblage dominated by colorless framework silicates.

2.2 Observations in XPL

Fragmental plagioclase mosaic. The field breaks up into a dense aggregate of angular plagioclase fragments in first-order grays, whites, and blacks. Many fragments display sharp polysynthetic (albite) twin lamellae; others are untwinned or are oriented near extinction. The fragments are angular to subangular, their boundaries are irregular and interlocking, and they "float" in a much finer-grained crushed groundmass of the same plagioclase. Several larger clasts show internal strain: undulose extinction sweeping across the grain, bent twin lamellae, and incipient breakup

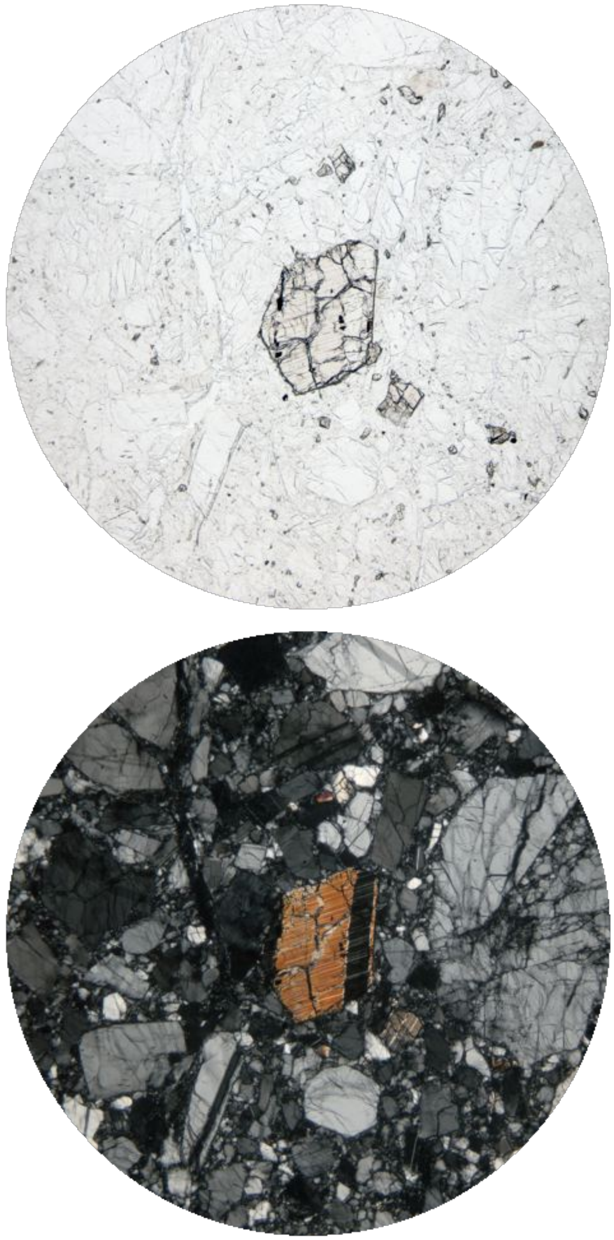

Figure 1: Sample 60025-r1 in PPL (top) and XPL (bottom). Field of view 2.0 mm

into slightly misoriented subdomains.

The central grain. In XPL, the central brown grain lights up in vivid second-order orange interference colors, far above the first-order grays of the surrounding feldspar — identifying it as a mafic silicate, olivine or pyroxene. A band along its right margin shows a contrasting dark extinction orientation with fine parallel lamellar structure; depending on orientation this may represent twinning or exsolution lamellae (which would favor pyroxene), though distinguishing olivine from pyroxene definitively would require checking extinction angles, cleavageon the microscope rather than from a single frame. The grain is itself heavily fractured, sharing in the deformation that shattered the host rock.

The rotation series shows the lamellar band is a twin, and that settles it. Watch the relationship between the orange body of the grain and the dark striped band on its margin as the stage turns. They extinguish at different stage positions — through the early frames the band is dark with fine bright lamellae while the body blazes orange; by the late frames the body has come back bright while the band sits dark. Two domains of one crystal alternating extinction across a sharp planar boundary, with fine parallel lamellae, is twinning — and in this context it's the classic (100) twinning of lunar clinopyroxene (pigeonite/augite), often combined with exsolution lamellae. Olivine cannot do this: it has no polysynthetic or lamellar twinning. This single observation eliminates olivine outright.

The extinction behavior is clean and inclined. The grain body sweeps smoothly from maximum brightness near frame 000 down to complete extinction around frames 060–070 (it's essentially black at 065), then brightens again toward 085 — sharp, uniform extinction confirming it's a single crystal domain, with the expected 90° periodicity. At the extinction position, the trace of the lamellae/long edge of the grain sits visibly oblique to the vertical and horizontal (the polarizer directions, assuming the standard N–S/E–W setup) — i.e., inclined extinction of very roughly 20° or more relative to the lamellar trace. Inclined extinction against the composition plane is clinopyroxene behavior; orthopyroxene would extinguish parallel to its cleavage/lamellae, and in any case opx is excluded by the interference colors.

Birefringence stays consistent with clinopyroxene. The body reaches bright second-order orange at maximum illumination. That's comfortably in augite/pigeonite range, too high for orthopyroxene (first-order grays only). So the verdict from the optics alone is a twinned clinopyroxene, most likely pigeonite or augite, which is exactly what the documented mineralogy of 60025 predicts: its mafic grains are augite and orthopyroxene with some exsolved pigeonite, while olivine is only a trace phase in mafic clumps generally absent from the thin sections. Note also the small grain at lower right shows the same twinned, alternating-extinction behavior — another clinopyroxene fragment [2].

Matrix. Between the recognizable clasts, the groundmass is a very fine, almost granular aggregate that appears mottled dark gray in XPL. There is no obvious isotropic glassy mesostasis; the matrix appears to be crystalline powder rather than impact melt, although a small amount of glass cannot be excluded at this scale.

2.3 Texture and Interpretation

The sample exhibits a coarse-grained, holocrystalline texture dominated by plagioclase. The interlocking grain boundaries and abundance of plagioclase indicate an anorthositic composition. Extensive fracturing and local brecciation textures suggest that the rock has experienced significant impact-related deformation after crystallization. Despite this deformation, the primary igneous plagioclase fabric remains largely preserved.

2.4 Interpretation

The texture is cataclastic: angular fragments of a coarse plagioclase-rich precursor sit in a matrix of their own crushed debris. Several lines of evidence support this reading. The clasts are angular rather than rounded or euhedral, so the fabric records mechanical deformation, not crystal growth. The seriate, fragment-in-powder size distribution is the signature of crushing in place. The pervasive undulose extinction and bent twins in the surviving larger clasts record plastic strain accompanying the brittle failure. And the mineralogical uniformity — nearly everything is plagioclase of similar character — suggests a single parent lithology (an anorthosite) rather than a mixed (polymict) breccia, though a full slide survey would be needed to confirm that no foreign clast types are present.

On the Moon, this texture is the fingerprint of impact processing. A coarse-grained ferroan-anorthositic crustal rock was shattered — likely repeatedly — by impact events, crushing much of it to angular debris while leaving relict crystals of plagioclase and a few mafic grains as survivors. The shock level indicated is moderate: enough to fracture, strain, and comminute the rock, but not enough to convert the plagioclase to maskelynite (diaplectic glass), which would appear isotropic black in XPL, nor to melt the matrix into a flowing glassy groundmass. The twinned, birefringent plagioclase fragments clearly retain their crystallinity.

The estimated mode from these images is on the order of 95% or more plagioclase [2], a few percent mafic silicate (the central grain and its smaller companions), and a trace of opaques — an anorthositic bulk composition. The presence of a millimeter-scale mafic grain suggests the precursor may have been an anorthosite with minor olivine or pyroxene.

3 Summary

The PPL/XPL rotational pairs shows a cataclastic anorthositic rock: angular, strained, polysynthetically twinned plagioclase clasts in a seriate distribution down to a finely comminuted plagioclase matrix, with a prominent fractured pyroxene at the center of the field and only trace opaques. The fabric records the impact shattering of a coarse-grained, plagioclase-rich crustal precursor — moderate shock, below the thresholds for maskelynite formation or matrix melting — and is typical of the brecciated ferroan anorthosites that dominate the lunar highlands collection.

References

[1] *The Virtual Microscope*, The Open University. [Online]. Available: `https://www.virtualmicroscope.org/content/60025-anorthosite-cataclasite`

[2] C. Meyer, *Lunar Sample Compendium*. Houston, TX, USA: NASA Johnson Space Center, 2011. [Online]. Available: `https://curator.jsc.nasa.gov/lunar/lsc/`